

# Proof of Stake on Sequentia

Consensus specification

**Abstract.** Sequentia is a UTXO blockchain for issued assets that subordinates itself to Bitcoin. Every Sequentia block commits to a Bitcoin block, and the chain reorganises whenever Bitcoin does, so a Sequentia transaction is never more settled than the Bitcoin history it is anchored to. This property exists for one purpose: it lets an asset issued on Sequentia be exchanged for bitcoin on the Bitcoin chain in real time, without the confirmation depth or protective timelocks that cross-chain trades otherwise require, because a Bitcoin reorganisation that reverses the bitcoin leg reverses the Sequentia leg by consensus rule. Block production is by proof of stake. A stake is an ordinary unspent output of the staking token, Sequence, and it never moves; a verifiable random function elects producers privately, in proportion to stake; a public committee of 250 certifies each block with one aggregate signature; and finality is layered, with quorum finality subordinated to Bitcoin checkpoints and both subordinated to the anchor. The protocol further expresses vesting schedules whose tokens stake while non-transferable, non-custodial delegation, and consensus-enforced reward distribution for staking pools. This document specifies the mechanism and states which of its guarantees are enforced by consensus and which rest on incentives and monitoring.

## 1. Introduction

---

Sequentia is a blockchain for issued assets built on the unspent-transaction-output model and Bitcoin Script. It derives from Elements, the Bitcoin sidechain codebase, from which it inherits its transaction format, its support for asset issuance, and its script extensions, Taproot and transaction introspection among them. Where Elements assigns block production to a fixed federation, Sequentia assigns it by proof of stake. The signed-block machinery is inherited unchanged: a block is valid only if it satisfies a challenge script fixed by consensus, and the rules of this document determine which keys that script names.

Three commitments govern the design.

**Bitcoin's history prevails.** Every Sequentia block commits to a Bitcoin block, and Sequentia reorganises whenever Bitcoin reorganises, immediately and unconditionally; no internal notion of finality can veto a reorganisation driven by Bitcoin. Section 2 develops this mechanism, anchoring, and the settlement guarantee it exists to provide; Section 6 places it above both of the protocol's finality layers.

**New assets may be issued at any time; new Sequence may not.** Sequentia carries assets created on it by their issuers, and asset issuance is an ordinary, ongoing use of the chain. Its staking token, named Sequence with ticker SEQ, is the exception: its supply of 400,000,000 units exists in full from the genesis block, no coinbase ever creates more, and there is no block subsidy. A block producer's entire income is the transaction fees of its block.

**Sequence is privileged only as stake.** In every other respect Sequence stands equal to the assets issued on the network: transaction fees are payable in any issued asset a producer accepts, and

holding Sequence is not a requirement for transacting. Its single privileged role is staking, and consensus admits only Sequence as stake, verifying the asset identifier explicitly.

Table 1 lists the protocol's constants. Every value in the table is a consensus rule of the network, identical on every node. None is a node setting: a Sequentia node whose configuration attempts to alter any of them refuses to start.

| constant                | value                                     | definition                                                                                                     |
|-------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| supply of Sequence      | 400,000,000 SEQ                           | issued in full at genesis; never increased                                                                     |
| minimum stake           | 40,000 SEQ                                | 0.01% of the supply; the least total stake with which a key participates in block production (Section 3)       |
| slot interval           | 30 seconds                                | the nominal block interval; also the unit of the production time gate (Section 4.2)                            |
| maximum block weight    | 200,000                                   | one twentieth of Bitcoin's, so that at the slot interval the chain's capacity per unit time equals Bitcoin's   |
| unbonding period        | 43,200 blocks                             | 15 days of slots; the least relative timelock of a staking output (Section 3)                                  |
| committee cap           | 250                                       | the certifying committee is the smaller of the cap and the number of eligible registered stakers (Section 4.3) |
| quorum                  | $\text{floor}(K/2) + 1$ , plus 1 at odd K | computed from the actual committee size K; 126 at the cap                                                      |
| escape anchor gap       | 3 Bitcoin blocks                          | the Bitcoin progress after which a stalled chain accepts a below-quorum block (Section 5.4)                    |
| checkpoint burial depth | 2,016 Bitcoin blocks                      | approximately two weeks; the burial at which a checkpoint finalises (Section 6.3)                              |
| payout notice period    | 2,880 blocks                              | 24 hours of slots; the least delay before a declared payout policy binds (Section 10)                          |

Table 1. Consensus constants of the Sequentia network.

The document proceeds from the outside in: the Bitcoin anchor and the settlement guarantee (Section 2), the stake (Section 3), the election of producers and certifiers (Section 4), agreement within a height (Section 5), and finality (Section 6). Sections 7 through 10 specify the economic layer: rewards, vesting, delegation, and reward distribution. Section 11 states the trust model.

## 2. Bitcoin Anchoring and Real-Time Cross-Chain Settlement

---

### 2.1 The anchor

Every Sequentia block names a Bitcoin block, its *anchor*, by height and hash. Each Sequentia node observes the Bitcoin chain through a Bitcoin node, and three rules bind the reference.

*Validity.* The anchor must be a block of the Bitcoin chain as observed by the validating node.

*Monotonicity.* A block's anchor height is not lower than its parent's, so the anchored view of Bitcoin only advances along a Sequentia chain.

*Canonicity.* When a Bitcoin reorganisation removes an anchor from Bitcoin's best chain, every Sequentia block carrying that anchor is invalidated at once, together with all of its descendants. The chain reorganises onto its most recent validly anchored block. If a subsequent Bitcoin reorganisation restores the anchor, the invalidated blocks are reconsidered and may rejoin the chain.

Canonicity is enforced continuously, not at a depth: a Sequentia block is discarded the moment its anchor is orphaned, whatever certification it carries (Section 6.4). Block production tracks Bitcoin closely from the other side as well: committee members presented with competing proposals rank them first by the recency of their anchors (Section 5.1), so the certified chain follows Bitcoin's tip.

### 2.2 The settlement guarantee

The purpose of the anchor is to let bitcoin and Sequentia assets be exchanged against each other in real time.

Consider the sale of an asset issued on Sequentia against a payment in bitcoin on the Bitcoin chain. Two assets issued on Sequentia trade atomically in a single transaction; the cross-chain trade cannot, because its two legs confirm on two chains, each of which can reorganise independently. The classical hazard is that one leg is reversed after the other has settled: the bitcoin payment is reorganised away while the asset transfer survives, or the reverse. Conventional practice defends each party with confirmation depth and with contract timelocks sized against reorganisation risk, and a cross-chain trade inherits hours of latency from those margins.

Anchoring replaces the margins with a consensus rule.

**Property 1 (settlement inheritance).** Let  $t_B$  be a Bitcoin transaction included in Bitcoin block  $b$ , and let  $t_S$  be a Sequentia transaction included in a Sequentia block  $s$  whose anchor is  $b$  or a descendant of  $b$ . In every evolution of the two chains in which  $b$  leaves Bitcoin's best chain,  $s$  leaves Sequentia's best chain.

The property follows directly from canonicity: a reorganisation that orphans  $b$  orphans every Bitcoin descendant of  $b$ , so the anchor of  $s$  ceases to be canonical and  $s$  is invalidated together with everything built upon it.

Its consequence is that the two legs of the trade can be made to fail together. The seller of the asset treats the trade as settled when two conditions hold: the block  $s$  containing the asset transfer is

quorum-certified, a matter of seconds (Section 5), and the anchor of  $s$  is at or above  $b$ . From that moment, any Bitcoin reorganisation deep enough to reverse the bitcoin payment necessarily reverses the asset transfer with it. The bitcoin payment must be included in a Bitcoin block; it does not need to be buried beneath further ones, because burial adds nothing that Property 1 does not already provide. The trade settles at the pace of one Bitcoin inclusion and one Sequentia certificate.

The same argument sizes the timelocks of hash-locked contracts. Payment channels and atomic swaps spanning two independent chains ordinarily pad their timelocks with a reorganisation margin on each chain. Between Bitcoin and Sequentia the padding serves no purpose: a reorganisation that undoes one leg undoes the other by rule, so timelocks are sized for protocol liveness, the time honest parties need to act, and for nothing else.

### 2.3 The anchor as a randomness beacon

The anchor serves a second role: it supplies the randomness of leader election. The election seed of every height is derived from the parent block's anchor (Section 4.1), so the unpredictability of the election rests on Bitcoin proof of work, which no Sequentia participant can bias or grind.

## 3. Stake

---

**Definition (staking output).** A staking output is an unspent transaction output satisfying three conditions: (i) its value and asset are explicit, and the asset is Sequence, since a confidential value cannot carry publicly verifiable weight; (ii) its script is exactly the template below, stored bare so that validators recognise it at creation; and (iii) its relative timelock, measured as wall-clock time, is at least the unbonding period. The *weight* of a staking output is its value.

```
<csv> OP_CHECKSEQUENCEVERIFY OP_DROP

    [ <bls_pubkey> OP_DROP <bls_pop> OP_DROP ]           committee registration
(Section 4.3)

    [ <liquid_locktime> OP_CHECKLOCKTIMEVERIFY OP_DROP ]   vesting lock (Section 8)

<pubkey> OP_CHECKSIG
```

The staking script. Bracketed fields are fields a staking output may omit.

The staker set is a pure function of the unspent-output set. Each node derives it by scanning that set at startup and maintains it incrementally as blocks connect and disconnect; carrying no state of its own, it is correct across reorganisations by construction. Weight aggregates per public key across all of a key's staking outputs, and the minimum stake of 40,000 SEQ applies to the aggregate, delegated weight included (Section 9). Keys below the minimum are excluded from leader election, from committee selection, and from the totals against which other stakers are measured.

Two consequences of the definition organise much of what follows.

First, staking involves no transfer, no module, and no custodian. Producing a block neither spends nor references the staking output (Section 4.2); the output confers weight by existing. Unbonding is the act

of spending the output, which the script's relative timelock delays by the unbonding period of fifteen days, and the weight lapses when the output is spent.

Second, weight is independent of spendability. An output that cannot at present be spent holds its full weight nonetheless. Section 8 builds vesting on exactly this independence.

## 4. Election

---

### 4.1 The seed

Each height  $H$  of the chain has an election seed

```
seed(H) = SHA256( anchor(parent) || H )
```

where  $\text{anchor}(\text{parent})$  is the hash of the Bitcoin block anchored by the parent. Every node computes the seed identically and without interaction. Because the anchor is a product of Bitcoin proof of work, no staker chooses it or predicts it, and the elections it seeds cannot be steered from within the network.

### 4.2 Producer sortition

The entitlement to produce is decided by private sortition. A staker with secret key  $sk$  and weight  $w$ , of total eligible weight  $W$ , evaluates a verifiable random function over the seed and reduces the output to a slot distributed uniformly on  $[0, W/w)$ . The staker is entitled to publish a block at height  $H$  from time  $T_{\text{parent}} + \text{slot} \times 30$  seconds, where  $T_{\text{parent}}$  is the parent's timestamp. The block carries the VRF proof, and every validator verifies that the proof corresponds to the staker's key and the seed and that the block's timestamp respects the resulting slot. Greater weight yields statistically earlier slots, so the frequency of production grows with stake.

The sortition is private: a VRF output is computable only with the secret key, so until a block is published no observer can identify who is entitled to publish it. There is no forward schedule to attack, bribe, or censor. Absence is harmless: if the earliest entitled staker publishes nothing, the next slot arrives thirty seconds later and entitles another.

### 4.3 Committee selection

Certification requires the opposite of privacy: validators must know whose signatures a certificate needs. Committee selection is therefore public and deterministic. For the seed of a height, every eligible staker receives a ticket

```
ticket(pk) = SHA256( seed || pk ) / w
```

with the hash read as a 256-bit integer and divided by the staker's weight. Stakers are ordered by ascending ticket, and the committee comprises the first  $K$  entries holding a registered committee key, where  $K$  is the smaller of the committee cap of 250 and the number of such stakers.

The committee key is a BLS12-381 public key. A staker registers it by embedding it, together with a

proof of possession, inside a staking output; the fields are inert with respect to spending, and the registry learns them from the unspent-output set exactly as it learns weight. The proof of possession is verified when the block creating the output connects, and consensus admits at most one committee key per staker.

A committee member certifies a block by signing its hash with its committee key. BLS signatures over a single message aggregate, so a block's certificate consists of a bitfield naming the signing members within the ordered committee and one aggregate signature: approximately three hundred bytes at any committee size, verified in a single aggregate check against the registered keys. The block itself is signed by its producer with the ordinary key of its staking output, as the challenge script requires; only certification uses BLS.

The quorum at committee size  $K$  is

$$q(K) = \text{floor}(K/2) + 1, \text{ plus } 1 \text{ when } K \text{ is odd}$$

which is 126 at the cap and is computed from the actual committee, so a network of sixty eligible registered stakers runs a committee of sixty with a quorum of thirty-one.

**Property 2 (quorum intersection).** For every  $K$ , any two quorums intersect in at least  $2q(K) - K \geq 2$  members. Two certificates for distinct blocks at one height therefore require at least two members to have signed both.

The two constants answer two different adversaries and are independent. The quorum is a bare majority, rather than the two-thirds familiar from Byzantine fault tolerance, because the binding constraint on the quorum is liveness under partial participation: a two-thirds quorum is unreachable whenever a third of the committee is merely offline. The size is 250 because size is what resists capture: a coalition holding a third of all stake occupies, by chance, a majority of a 100-member committee within days, and a majority of a 250-member committee on the order of once in decades. Size defends safety against a hostile third of the stake; the majority quorum defends liveness against an absent third of the committee.

Property 2 makes double certification attributable only if no member signs two proposals at one height. The round structure does not by itself guarantee that of honest members; Section 5.3 states the rule that does.

## 5. Certification

---

### 5.1 Rounds

Time at a height is divided into rounds, measured from the timestamp of the first proposal. In the opening portion of a round, the collection window, members gather the proposals that have arrived rather than endorsing the first. When the window closes, every member backs the same proposal, chosen by a ranking every node evaluates identically: proposals carrying more recent Bitcoin anchors rank first, and among equal anchors the lower producer VRF value wins. In the common case all proposals carry the same anchor and the ranking reduces to pure lowest-VRF election; the anchor

term binds only when a producer anchors stale, and it is the committee-side half of the freshness guarantee of Section 2. Each member backs by flooding a BLS signature share over the block hash; any node that accumulates a quorum of shares assembles the certificate and floods it. If the round ends without a certificate, the next round begins at the same height and members back the next proposal in the same ranking, so a silent producer costs one round.

The collection window and the round last  $500 + 25K$  and  $700 + 35K$  milliseconds respectively, 6.75 and 9.45 seconds at the committee cap, growing with  $K$  because share traffic and share verification are the only per-member costs. These are liveness constants, identical on every node, and they are not validity conditions: no block is accepted or rejected by reason of a clock. Certification proceeds at the pace of the fastest quorum, since any single node holding a quorum of shares completes the certificate, and the slowest members contribute nothing to latency. A mistimed constant therefore costs delay, never disagreement; the one respect in which timing once touched safety is removed by the share-lock of Section 5.3.

## **5.2 The certificate as an independent object**

A certificate is self-verifying: hash, bitfield, and aggregate signature, checked against the registered committee keys. It is flooded as an object in its own right, separately from the block body, which may be three orders of magnitude larger.

The separation is a safety mechanism. If a partition swallows the block body while the certificate crosses, a node beyond the partition that holds the certificate knows the height is decided: it pins its finality floor to the certified block, requests the body, and neither produces nor countersigns a rival at that height, and the escape valve of Section 5.4 is inhibited there. Without the certificate as an independent object, the far side of a partition could build upon a height that had already been decided against it.

## **5.3 The share-lock**

Rounds create one honest path to conflicting certificates. Suppose block  $X$  gathers its final share shortly before a round boundary. Some node assembles the certificate and begins flooding it, but flooding takes time, and the boundary passes first. Members that have not yet received the certificate treat the round as failed and back the following round's proposal at the same height. The shares they issued for  $X$  remain valid, so  $X$ 's certificate completes regardless; if the second proposal also reaches quorum, two certificates exist at one height, and every signature in both was issued honestly, one round apart, exactly as the clock instructed.

The share-lock closes the path. A member that has issued a share for  $X$  at height  $H$  issues no share for any other block at  $H$ , and does not extend a chain that excludes  $X$  above  $H$ , until  $X$  can no longer certify. The lock releases when three conditions hold: the round in which the member signed has ended; the member has queried its peers for a certificate on  $X$ , a three-hundred-byte round trip; and one further round has elapsed without an answer. Where a partition may be concealing the certificate, so that no query can settle the question, release is governed instead by the escape gap: the member holds until the Bitcoin chain has advanced three blocks past the parent's anchor.

**Property 3 (certificate uniqueness).** While the lock holds, at most one certificate can exist at a height: for  $X$  to remain capable of certifying, at least  $q$  members must have signed it, and all of them are locked, so fewer than  $q$  members remain free to sign a rival, and no rival reaches quorum.

In routine operation the lock binds nothing, since certificates complete promptly and release it at once; its cost is one round of delay behind a failed proposal. It is also what disarms the timing constants of Section 5.1: with the lock in place, a race at a round boundary yields a delayed certificate rather than a second one.

## 5.4 The escape valve

A committee too depleted to reach quorum would otherwise stall the chain indefinitely. A block certified by fewer than the quorum, at minimum by its producer alone, is accepted under exactly one condition: its anchor lies at least three Bitcoin blocks beyond its parent's, approximately thirty minutes of Bitcoin work. Liveness is thereby restored on Bitcoin's clock, without relaxing the quorum in normal operation. A below-quorum block never advances the finality point of Section 6.1, ranks beneath any quorum-certified competitor, and is not produced at heights where a certificate is known.

## 5.5 Ordering competing blocks

A node orders competing blocks at a height by three keys: certification weight first, so a quorum-certified block outranks everything else; the producer's VRF value second, lower winning; and the block hash third, lower winning. The final key is globally deterministic, so nodes that have observed the same candidates order them identically, and a symmetric tie between below-quorum siblings resolves the same way everywhere without coordination.

No key of the ordering, and no other rule of the protocol, reorders a block that is certified. Certification is never revisited on account of height, arrival rate, or anchor freshness: a protocol that revisited it could be outrun by an adversary manufacturing certified blocks faster than the honest chain produces them.

# 6. Finality

---

## 6.1 Immediate finality

The highest quorum-certified block of a node's active chain is its finality point. Blocks at or below the finality point are not reorganised by any Sequencia-internal cause; competing branches are rejected at acceptance. So strong a rule is affordable because of Properties 2 and 3, which together ensure that a second certificate at a finalised height cannot exist unless two members equivocate, an act their own published signatures prove.

Immediate finality protects a participant whose node was online to observe the certificate. It says nothing to a node that was absent, and nothing about histories fabricated after the fact, which are the subject of the next two subsections.



## 6.2 The long-range problem

Stake secures a chain only while it is bonded. Keys that once held a majority of stake, and have since unbonded and divested, can sign an alternative history branching from any point of their tenure, at no cost and with valid signatures throughout. A node synchronising from scratch observes two internally consistent histories and has no intrinsic criterion for preferring the genuine one. Every proof-of-stake system requires an external criterion. Sequentia's is Bitcoin.

## 6.3 Checkpoints

Anyone may commit a Sequentia block reference into Bitcoin: a transaction output carrying a fixed tag, the block hash, and the height. A node treats the referenced block as checkpoint-finalised when two conditions hold: the block lies on the node's own validated active chain, and the committing Bitcoin transaction is buried 2,016 Bitcoin blocks deep, approximately two weeks. Forks at or below the checkpoint-finalised height are rejected outright.

Checkpoints are reject-only. A checkpoint never causes a node to fetch, prefer, or adopt a chain; it converts history the node has already validated from final-by-quorum into final-by-Bitcoin. A checkpoint referencing a block the node has not validated is inert, so forged or conflicting checkpoints have no effect. A node deployed for the first time, which has validated nothing, is provisioned with a static checkpoint, a height and hash pair distributed with the software release, and applies it with the same reject-only semantics.

**Property 4 (posterior-corruption closure).** The unbonding period exceeds the checkpoint burial window: fifteen days of slots against 2,016 Bitcoin blocks, approximately fourteen days in expectation. Stake that certified a history therefore remains bonded until the checkpoints over that history have consolidated, so a coalition cannot unbond, divest, and afterwards rewrite the history it signed. Beneath every checkpoint, rewriting Sequentia requires rewriting two weeks of Bitcoin proof of work.

## 6.4 Precedence

The three notions of settlement stand in strict order: the anchor outranks checkpoints, and checkpoints outrank immediate finality. If a Bitcoin reorganisation orphans the anchor of a Sequentia block, that block is discarded whether it was certified, finalised, or beneath a checkpoint, and the finality point recomputes on the surviving chain. A node that discards a finalised block on this account is complying with the protocol.

The order is forced by Property 1. The guarantee extended to a cross-chain counterparty is precisely that a Sequentia settlement never outlives the Bitcoin history it is anchored to; an internal finality rule empowered to veto an anchor-driven reorganisation would break that guarantee in exactly the circumstances for which it is bought.

## 7. Producer Rewards

---

No block creates new Sequence, so a block's reward is the fees of its transactions, in whatever assets they were paid. Consensus fixes the destination: every coinbase output carrying value must pay the

required payee of the height, a script  $R(H)$  determined by the rules of Section 10, and a block whose coinbase pays anything else is invalid. Where the producer has declared no distribution policy,  $R(H)$  is a pay-to-witness-public-key-hash of the producer's key.

The producer constructs its coinbase by evaluating the same function that validators enforce, so the two cannot diverge. Rewards are paid in ordinary outputs distinct from any staking output: income from staking is transferable immediately, whatever locks encumber the stake itself.

## 8. Vesting and Staking-Only Periods

---

Allocations of Sequence distributed at genesis vest under schedules comprising a cliff, a linear vesting period, and a *staking-only period*: an interval in which the tokens can stake but cannot yet be transferred. The staking-only period rests on the independence of weight from spendability established in Section 3.

A staking output may carry an absolute timelock, the `liquid_locktime` field of the template. Until the stated time the output cannot be spent, and therefore cannot be sold, transferred, or pledged; throughout, it holds its full weight, produces blocks when elected, and earns fees, which are paid outside the output and are immediately transferable. Non-transferability is obtained as non-spendability, with no covenant, no escrow, and no custodian.

The lock must be absolute rather than relative. Relative timelocks are sixteen-bit quantities, bounded by 388 days in their time encoding and by 22.75 days in their height encoding at the slot interval, so a staking-only period of years is not expressible relatively. Absolute locktimes are thirty-two-bit and reach any date; they are expressed as wall-clock times, because block heights drift against the calendar over the length of a vesting schedule.

The cliff, during which an allocation must hold no weight at all, is obtained by not yet creating a staking output: the allocation waits in a plain timelocked output, which confers nothing, and is converted into its staking output when the cliff ends. Where the conversion itself must be trustless rather than merely public, the pre-cliff output is a Taproot covenant admitting exactly one destination, the staking script it is due to become; the destination is a constant fixed when the allocation is created, so the covenant is a hash comparison rather than a recursive construction.

## 9. Delegation

---

The key of a staking output performs two roles: it is the identity elected to produce, which must be online continuously, and it is the key that spends the output. Housing both roles in one key would place the key that controls the coins on an exposed machine, and, for a vesting-locked output that cannot be respent for years, would make the block-signing identity irrevocable for the term of the lock. Delegation separates the roles.

A staker, the *controller*, assigns its production rights to a *signer* by funding a delegation record, a bare output of the form

```
<"SEQDEL"> OP_DROP <signer> OP_DROP <controller> OP_CHECKSIG
```

spendable by the controller alone. While the record is unspent, the controller's aggregate weight counts toward the signer in producer sortition, in committee selection, and against the minimum stake; the staking outputs themselves are untouched, and their key is thereby freed to remain offline. Resolution is a single hop: weight delegated to a signer that has itself delegated does not follow the second edge, which forecloses cycles. Consensus admits at most one unspent record per controller, since a second would make the identity entitled to produce with that weight ambiguous; a block creating a duplicate is invalid.

Three properties follow.

*Custody is never conveyed.* The signer appears nowhere in the staking output's spending condition. A signer, honest or hostile, cannot move one token of the stake it produces with.

*Exit is unilateral and immediate.* The controller revokes by spending the record, an act requiring nobody's cooperation, incurring no delay, and moving no staked coin. Re-assignment spends the old record and creates the new one in a single transaction.

*Assignment survives the vesting lock.* The record is a separate output, so revoking or re-pointing the production rights of a locked stake never requires spending the stake. A compromised signer is replaceable at once, years before the coins can move.

Delegation is the protocol's staking-pool primitive. Holders below the minimum stake, or unwilling to operate infrastructure, assign their weight to an operator. The operator's income, and its obligations to its delegators, are the subject of Section 10.

## 10. Reward Distribution Policies

---

Delegation lends weight, and under Section 7 the rewards it generates accrue wholly to the pool operator, leaving delegators dependent on the operator's bookkeeping. A producer may instead bind itself to a distribution policy, declared in a payout record, a bare output of the form

```
<"SEQPAY"> OP_DROP <activation> OP_DROP <mode> OP_DROP <parameter> OP_DROP <signer>  
OP_CHECKSIG
```

spendable by the signer alone. The policy in force for a producer at height  $h$  is its record of greatest activation not exceeding  $h$ . The required payee  $R(h)$  of Section 7 derives from that policy, and consensus forbids two unspent records sharing one activation height for one signer. Two modes exist.

**Direct.** The parameter is a script, and  $R(h)$  is that script: the coinbase must pay the committed destination, and the operator cannot redirect a reward elsewhere. The rule constrains the destination, not the distribution; the protocol does not evaluate whether the destination is fair to anyone. Direct mode is a public commitment with consensus force, not a trustless split.

**Lottery.** The coinbase must pay a single participant, drawn from all stakers whose weight counts toward the producer, the producer's own stake included, with probability proportional to weight. The draw is a deterministic function, by domain-separated hashing, of the height's election seed, so every

node computes the same payee and a block paying any other is invalid; the seed being Bitcoin's, no participant can bias the draw. The parameter is a commission in basis points: with probability  $c/10,000$  the reward goes to the producer outright. Lottery mode pays each participant its exact proportional share in expectation, with no accounting subsystem. What it does not do is smooth the income: a participant holding a share  $p$  of the pool's weight receives the whole of a block's reward with probability  $p$ , rather than a share  $p$  of every reward.

Every policy is subject to the notice period: a record's activation must lie at least 2,880 blocks, twenty-four hours of slots, beyond the block that creates it, and a block violating the bound is invalid. The notice is what turns inspection into protection. A delegator's exit is immediate and unilateral (Section 9), so a policy change that must stand public for a day before taking effect cannot pay its operator anything before every delegator has had the opportunity to leave; absent the notice, an operator could accept delegation under one policy and harvest the next block under another. A declaration does not cancel a standing policy: through the notice window the standing record and the pending one coexist, the standing one in force.

## 11. The Trust Model

---

The guarantees of this specification divide cleanly by their enforcement.

**Enforced by consensus.** A settlement anchored to Bitcoin history cannot outlive that history, and history beneath a checkpoint cannot be rewritten without rewriting two weeks of Bitcoin proof of work (Properties 1 and 4). A vesting lock cannot be circumvented by its own holder. A pool operator can never spend a delegator's coins, and can never prevent, delay, or condition a delegator's exit. A declared payout policy cannot take effect with less than a day's public notice. Under lottery distribution, expected income is exactly proportional to weight, and the draw cannot be biased by any participant.

**Not enforced by consensus.** Direct distribution does not make an operator equitable: the committed destination is stable and public, not fair. The notice period protects the delegator who observes it; the protocol publishes the pending change and cannot compel anyone to look, so its practical force depends on wallets and explorers surfacing pending policy changes to the delegators they affect. Lottery income is exact only in expectation, and a participant requiring smooth income must either accept the terms of a direct-mode operator or hold stake sufficient to produce in its own right. An operator without stake of its own has no staking output in which to register a committee key: it can produce blocks with weight lent to it, but to countersign it must bond stake of its own.

The position taken by the design is that custody, exit, and commitment are enforceable, and that fairness beyond them is a market outcome: operators compete under policies that cannot be changed quietly, before delegators who cannot be detained.

## 12. Summary

---

A stake on Sequentia is an unspent output of Sequence that never moves, conferring the right to be elected in proportion to its value. Bitcoin supplies the election's randomness through the anchor each

block carries, and the anchor subordinates the whole chain to Bitcoin's history, which is what lets bitcoin and Sequentia assets change hands against each other in real time. A block is produced by a privately elected staker, certified by one aggregate signature of a public committee of 250, final upon that certificate, and immovable beneath a Bitcoin checkpoint, always excepting the anchor rule, which outranks both. Because weight is independent of spendability, tokens under vesting locks stake and earn while non-transferable. Because production rights are assigned in a revocable record separate from the stake, delegation conveys no custody and exit is immediate. And because a producer's reward destination is a consensus rule bound to its declared policy, a pool's delegators stand behind a commitment their operator can neither break quietly nor change quickly.

---

Sequentia is developed by Concatena Labs. The network is named Sequentia; the token used for staking is named Sequence, with ticker SEQ.